Predicting Marginal Win Value Using Baseball Team Statistics

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I. Introduction

For the 2023 Major League Baseball (MLB) season, the average revenue for an MLB team was $378 million (Ozanian and Teitelbaum 2024). Baseball is often called America's favorite pastime because of its storied history, and it has grown significantly since its humble beginnings in the late 1800s. Baseball history can be divided into different eras that mark changes in the way the game is played and how teams approach building their rosters. From 1900 – 1919, the Dead Ball era occurred, which is known for its low scoring games and dominant pitching. The Steroid Era occurred from 1994 – 2005 and is characterized by high scoring games and dominant hitting (Whalen, Matthews, and Mills 2024). As changes in league performance occur, teams constantly look for ways to improve. Elite starting pitchers[1] are more valuable to teams in eras where batters are dominant than in an era where the average scoring is low. As adjustments to team building occur, player compensation and the cost of running a franchise have increased. Shohei Ohtani received a ten-year, 700-million-dollar contract in the 2023 offseason. However, in 1980, the largest free agency contract offer was given to Dave Winfield for 10 years and 23-million-dollars (Randhawa and Kelly 2023). Going back slightly farther, yearly salaries above $1 million were not considered reasonable in the early 1970s (Scully 1974). Using data from the Minneapolis Federal Reserve, the value of $23 million in 1980 would be worth just under $88 million in 2023 ("Inflation Calculator | Federal Reserve Bank of Minneapolis," n.d.). This shows that the value of these contracts far outpaces the inflationary effects that money has experienced. As these contracts become more expensive and team payrolls continue to rise in both nominal and real terms, it is important that teams can

[1] Starting pitchers begin the game and pitch the majority of innings, typically 5-7 out of a 9-inning game. They are then taken out of the game and a relief pitcher is put in. Relief pitchers typically only pitch one or two innings before being taken out for a new relief pitcher.

accurately evaluate performance to ensure they are making quality investments with their resources ("Cot's Baseball Contracts," n.d.). As computers have been developed to handle large amounts of data, baseball fans and teams have begun analyzing the game they love. Similar to other sports, teams with higher payrolls typically are viewed as the favorites, but teams with smaller budgets look for ways to gain a competitive advantage. These small budget teams have begun to build their rosters based on new measures of performance and found success.

A unique difference between MLB and the other three major North American sports leagues is the absence of a strict salary cap. Typically, teams in professional sport leagues cannot spend more than a strict limit determined by the league. In the MLB, in place of a salary cap, there exists a Competitive Balance Tax, or luxury tax, to restrict excessive spending by the wealthier teams. Rather than there being a limit to spending, there is a penalty when a team is over the payroll threshold. The penalty is a percentage of this overage and is paid to the league, which then distributes it to the teams below the threshold. There is a union contract called the Collective Bargaining Agreement (CBA) between the teams and the Major League Baseball Players' Association (MLBPA) that stipulates the threshold for each year, as well as other regulations and payroll information. For 2024, this threshold was set at $237 million. The penalty rate depends on how much and how long the team has been over the threshold. If 2024 was the team's first season in the penalty zone, 20% of the total above $237 million was owed to the league. The second year over the threshold would lead to a 30% penalty, and if a team had been over for three or more years it carried a 50% penalty. The penalties for the size of the overages for 2024 are as follows: 12% for going $20 million over the threshold, 42.5% for going $40 million over for the first year and 45% each additional year, and 60% for going over $60 million ("Competitive Balance Tax | Glossary | MLB.Com," n.d.). This "soft cap" was

implemented to put a limit on the spending of richer teams while still allowing them to use the resources available to them, which maintains a semblance of competitive balance. This system necessitates teams with fewer resources to spend wisely because richer teams have no cap on their spending and can outbid teams for top players.

Baseball analytics came to the center of every baseball fan's attention when Michael Lewis wrote his book, *Moneyball*, which was later adapted into a film. In the book, he describes how the early 2000s Oakland Athletics built their team using new processes and analytics (Lewis 2004). In the late 1980s, the Athletics were one of the richest teams in the league; they would spend large amounts of money on free agents and give large contracts to homegrown players to keep them from leaving. In doing so, they made it to the World Series three years in a row. The team was then sold, and the new owner wanted to maintain competitiveness while reducing spending. The front office[2] took this challenge and found a way to gain a competitive advantage by spending less money on players undervalued by the market. Using this strategy, they were able to compete with teams who spent three times as much, like the Yankees (Lewis 2004). The analytics that these teams cared about were different from the mainstream statistics that most major league teams used. This new process for determining player value was given the term sabermetrics and soon began to spread to other teams throughout the league ("A Guide to Sabermetric Research – Society for American Baseball Research," n.d.). The goal of this paper is to determine the relationship between sabermetrics and different forms of team performance and if the advanced baseball statistics are better measures than the more dated baseball statistics.

[2] The front office is the group of people that control the operation of the organization, both the baseball and the business sides.

II. Sample

For the analysis, 1158 team observations are used from 1985 – 2024. This is the total number of teams that played a season over this time. 1985 was chosen because it is the earliest year that reliable payroll data was found for the Major League Baseball teams.

III. Dependent Variable

Marginal Payroll / Marginal Wins (Marginal Value of a Win)

Doug Pappas, a New York lawyer who conducted baseball research, proposed a measure of financial efficiency. It looks at how much a team spends above the minimum payroll for each additional win above a 0.300 winning percentage[3] (Pappas 2004). The original equation is:

$$MVW = \frac{Team\ Payroll - (28 * Major\ League\ Minimum)}{Games\ Played * (Winning\ Percentage - 0.300)}$$

Each year the major league minimum contract value is adjusted according to the CBA (as discussed previously) and is the minimum amount that an MLB player can make while being on a team's major league roster. The numerator in this equation is the amount that the team spends above the minimum amount required to fill a roster, which is composed of twenty-eight players. This minimum contract value player is often called a replacement level player. Due to the variability and randomness of baseball the projected minimum winning percentage for a team composed completely of replacement level players would be 0.300. The denominator finds the difference between the team and the replacement level team's winning percentage and multiplies

[3] The paper will use baseball terminology which denotes winning percentage as a decimal, so 0.300 for winning 30% of their games, 0.500 for 50%, and so on.
A 0.300 winning percentage is about 49 wins over the course of a 162 game season.

it by the number of games played in each season[4]. This statistic as a whole can be viewed as the marginal cost of a win better than the "worst case" scenario. Interpreting this statistic, if a team uses their resources efficiently, the marginal value of their wins will be lower than the league average. There are two ways to be efficient, either win more with the same budget as another team or win the same number of games with a smaller budget. A team can be better than the league average by either the numerator being smaller, or the denominator being larger than the league average. In both scenarios, teams will have a lower marginal win value when compared to the league average. This makes marginal win value a good measure for how smaller budget teams compete against larger budget teams.

The limitations of this statistic within the sample must be examined as well. Only four teams have won less games than the theoretical worst case scenario of a 0.300 winning percentage team since the inception of the amateur draft in 1965. The teams are the 2003 Detroit Tigers, the 2018 Baltimore Orioles, the 2019 Detroit Tigers, and the 2024 Chicago White Sox. For these four observations, the marginal win value for their season will be equal to the numerator in the Marginal Value of a Win equation. The difficulty with using this variable as a comparison over different years is that the league minimum salary changes over time as the CBA gets renegotiated every few years[5]. This creates a problem because as contract values increase so will the average marginal value for each win across the league. To mitigate this problem, this study will compare the marginal value of wins number for each team to the league average marginal value each year. This comparison statistic will be known as Marginal Win Value Plus (MWVP). This would mean that a team's performance is related directly to how they outperform

[4] Typically, this number is 162, but some teams play 161 or 163 depending on cancellations and tie-breaking games. In the shortened 1994, 1995, and 2020 seasons, the number of games played is used rather than 162. These totals were varied between 112 and 117 for 1994, 144 for 1995, and 60 for 2020.
[5] Refer to Appendix 1 for a graph of league minimum salary over time.

the other teams that year, which means they are making better decisions with the same information. Even though these difficulties exist, MWVP is the chosen main dependent variable because it shows that teams are making efficient and optimal financial choices on which players to sign and extend, rather than spending money with the hope that the most monetarily expensive players will lead to the most success. Teams with small or large payrolls can have a low marginal value of a win; teams with larger payrolls just need to win more (Pappas 2004). This statistic is attempting to achieve ceteris paribus by negating the effect that large payrolls have on success when determining how efficiently a baseball organization is being run. This most directly relates to the goal of the research by discovering if teams make more informed decisions. The salary data and win totals used to calculate this were both obtained from the baseball reference website[6].

In the regressions, the MWVP will undergo a log transformation. This will improve model interpretability because the MWVP statistic will be scaled to the league average so the coefficients in the model can be interpreted as the percent changes in comparison to the league average. Additionally, this log transformation will then be multiplied by –1, causing the higher values of the dependent variable equate to better team efficiency. These two changes will make the study easier to interpret and provide directly applicable results to how teams can outperform the league. The basic marginal value of win statistic is the amount of money that a team pays for each win above a 0.300-win percentage. A result of $5 million would mean that a team pays $5 million for each win above the 0.300-win percentage. For analysis, this value is divided by the league average and then multiplied by 100. A value of 125 at this stage means that a team's marginal win value is 25% higher than the league average and a value of 75 would mean that a

[6] Main website is www.baseballreference.com ("MLB Stats, Scores, History, & Records," n.d.)

team is 25% lower than the league average. The team with a value of 75 would be better at using their resources efficiently compared to the league average or the team with a value of 125 because they are paying less for each additional win by the team[7]. From here the number undergoes the log and negative transformation for an ease of interpretability. Before the negative transformation, the more efficient teams would be below the league average, but after it would be above the league average (i.e. 50 is lower than 100 but -50 is larger than -100). Therefore, during interpretation a decrease/increase in an independent variable will lead to a decrease/increase in efficiency. In the regressions, this transformed MWVP stat will be called the dependent variable for analysis, or DEPANA[8]. An increase in efficiency means that MWVP decreases, meaning that the team is performing better than the league average; a decrease in efficiency means that MWVP increases meaning that the team is performing worse than the league average.

IV. Independent Variables

Major League Statistics

Batters with a high batting average (BA)[9] and pitchers with a low Earned Run Average (ERA)[10] reigned supreme in the early years of baseball. These, along with other statistics and the "eye-test"[11], molded the way that general managers and owners built their baseball teams and invested money. As time has gone on, statisticians like Bill James, Tom Tango, Michael G. Lichtman, and Andrew E. Dolphin have called into question these methods and looked at what

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[7] See Appendix 2 for an in-depth analysis of the 2024 season regarding these variables.
[8] DEPANA and efficiency will be used interchangeably.
[9] BA Equation: $\frac{Hits}{At-Bats}$, the batting average is the percentage of the time a batter gets a hit in decimal form.
[10] ERA Equation: $\frac{Earned\ Runs}{Innings\ Pitched} * 9$, since the games are 9 innings long, this equation calculates the average number of runs a pitcher allows over an entire game.
[11] The "eye-test" means that scouts look at the way a player looks when they play baseball, i.e. do they have a good-looking swing and are they smooth in the field? Since this is completely subjective, scouting was viewed as an art of baseball and only former players could truly succeed in it because they knew what to look for.

truly leads baseball teams to success. These are some of the people who developed sabermetrics, which was previously discussed. This study uses both old era statistics, which is the “eye-test”, and new era statistics, which are sabermetrics, to determine if there is a stronger relation between either of the methods and team success. The data are from the baseball reference website and contains the years 1985 – 2024, which was scraped from the webpages using Python and the Beautiful Soup functionality. A year variable was added during this scraping and the statistics from eight different webpages were used: standard batting, advanced batting, value batting, sabermetric batting, standard pitching, advanced pitching, value pitching, and standard fielding. These were then compiled into a single dataset with each team for each year being a single observation. This makes the study applicable as teams can make informed decisions about which statistics to value when building the roster. Most of these variables should be correlated with both the numerator and the denominator of the marginal win value statistic because teams that perform better statistically should win more baseball games, but players who perform well cost more money. This study is looking to find which statistics have two different relationships with the numerator and the denominator. This would lead to either a positive or negative relationship with MWVP depending on which part of the equation the statistics have a greater relationship with. These are the main variables that will provide a relationship for team/organizational success. Below is a table listing the statistics that will be used, how they are calculated, if they are classified as an old era or new era statistic (for the purpose of this paper), and how they are hypothesized to affect the DEPANA variable. Pitching, Batting, or Fielding was added to the beginning of each of the variable names for ease of interpretability. This will make it easy to determine what facet of the game the statistic is trying to measure.

Table 1 Listed are the baseball statistics used as independent variables for analysis and the predicted effect on the outcome variable. Statistics, formulas, and analysis from (Tango, Lichtman, and Dolphin 2006; Castrovince 2020; Lewis 2004). As a reminder, a high MWVP means a less efficient use of resources while a low MWVP means a more efficient use of resources in relation to the league average. The regressions use a negative transformation of MWVP so that an increase in negMWVP is an increase in efficiency.

Variable	Explanation of Statistic	Era	Hypothesized Effect on efficiency (DEPANA) in relation to the league average when Variable Increases
Batting.3B	When a batter hits[12] the ball and gets to third base off of their hit alone. Known as triples.	Old	Increase, occurs the least out of all hits so money may not be given to it as much as other statistics.
Batting.SB	When a runner advances to the next base during a pitch, without the batter hitting the ball. Known as stolen bases.	Old	Decrease, stolen bases do not give enough value given the risk, but they often get players a lot of money.
Batting.CS	When a catcher throws the ball after a pitch and gets the runner who is trying to steal out	Old	Decrease, when teams get caught stealing a lot, they lose base runners and this negatively impacts wins.
Batting.BB	Occurs when the pitcher throws four pitches out of the strike zone that the batter does not swing at. The batter then is awarded first base instead of the at-bat[13] continuing. Known as walks or bases-on-balls.	Old	Increase, as a team walks more, they win more.
Batting.OPSP	(On-Base Percentage[14] + Slugging Percentage[15]) / League Average OPS[16]	New	Increase, measures offensive performance in relation to the league average. Effect on performance will be greater than effect on payroll so MWVP will decrease.

[12] A hit is classified as when the batter hits the ball and he advances to at least first base safely.
[13] At-bats is the number of times the player either got a hit or got an out. It excludes walks, hit by pitches, and sacrifice outs.
[14] On-base percentage is the total number of times that a player gets on base / the total number of opportunities the player has to get on base.
[15] Slugging Percentage is the total bases that a players earns, (1*Singles+2*Doubles+3*Triples+4*Home Runs)/At-bats. Singles are when the batter gets 1 base off a hit. Doubles occur when the batter gets 2 bases. Triples occur when the batter gets 3 bases. While home runs are when the batter hits the ball over the outfield wall. Each multiplier is the number of bases the player gets to safely for each type of hit,
[16] The same calculation done in the numerator is done in the denominator, but for the league rather than the player.

Batting.SO	When a batter gets out without putting the ball into play. Occurs when three pitches are either: thrown into the strike zone, the batter swings and misses, or the batter hits a foul ball during an at-bat (cannot end on a foul ball).	Old	Decrease, teams that strikeout more typically underperform for how much money they make.
Batting.TB	(1*Singles+2*Doubles+ 3*Triples+ 4*Home Runs) This is also the numerator in the slugging percentage calculation.	New	Increase, this measures total offense and as a team's offense gets better so will the team's chances at winning.
Batting.GDP	Grounded into double plays. When one ball in play leads to two outs for the fielding team.	Old	Decrease, double plays end innings very quickly because 1 play gets the defense 2/3 of the necessary outs to end the inning.
Batting.HBP	When a pitch hits the batter, the batter is awarded first base without the at-bat continuing.	Old	Increase, mostly random, does not have to do with the skill of players, but are just as valuable as a walk so will increase odds of winning.
Batting.SH	When a batter hits a ball in play and a runner already on base advances, but the batter gets out.	Old	Decrease, this is typically a strategic play, trading an out for an additional base or run, but some previous studies have brought into question the effectiveness of this strategy as a long-term winning strategy.
Batting.SF	When a batter hits a ball in the air and the ball is caught, but a runner already on base advances.	Old	Increase, typically this is exchanging an out for a run, which is more effective than the sacrifice hit described above. Usually this shows that a team is aggressive, which is helpful in winning games.
Batting.rOBA	Weighted On-Base Percentage. This is calculated similarly to On-Base + Slugging (OPS), but it gives a weight to every way you get on base[17].	New	Increase, a good metric for measuring the effectiveness of an offense holistically. When the offense performs better so will the team in terms of winning games.

[17] (BBf*BB + HBPf*HBP + 1Bf*1B + 2Bf*2B + 3Bf*3B + HRf*HR)/(At-Bats + BB + SF + HBP). The variables with an f on the end are the weights, which are determined by the analysts based on the values provided by each outcome each year.

Batting.BAbip	Batting average on balls in play. Only balls that are put into play count for this statistic. (H-HR)/(AB – SO – HR + SF).	New	Increase, typically viewed as a luck statistic if especially high then the offense is very lucky, while if very low the offense is unlucky. This means that as this increases the team is overperforming their normal level and therefore overperforming given their pay.
Batting.ISO	Slugging Percentage – Batting Average[18]	New	Increase, this measures an offense's ability to hit for power. More extra base hits are very valuable so as this increases, so should a team's wins.
Batting.BA	Hits/At-bats	Old	Decrease, due to the inefficiency in providing accurate analysis using this statistic. Leads to the prediction that there will be a significant, negative relationship.
Batting.HRper	HR/all plate appearances[19]	Old	Increase, as a team hits more home runs, their chances of winning should increase.
Batting.SOper	SO/all plate appearances	Old	Decrease, as a team strikeouts more their chances of winning should decrease.
Batting.BBper	BB/all plate appearances	Old	Increase, as a team walks more their chances of winning should increase.
Batting.RSper	(Runs - HR) / (Hits + HBP + BB – HR + G_pr[20])	New	Increase, when an offense is more efficient at converting baserunners to runs, they should win more often than teams who are inefficient at this.
Batting.SBper	SB / (SB+CS)	Old	Increase, stolen bases add value but only when taken in reference to how successful they are. If a team gets caught stealing often then it cancels out the value added by stealing the base.
Batting.XBTper	The percentage of time a baserunner takes more than	Old	As baserunners are better at getting more bases for each hit,

[18] See Batting.BA
[19] Plate appearances count all times that the batter had the opportunity to put the ball in play. It includes at-bats and the three statistics mentioned in footnote 13 that at-bats don't include.
[20] Games played as a pinch runner. Pinch runners are people that are substituted in for a hitter who got on base.

	1 base on a single or 2 bases on a double. This could occur because of a bad throw from a fielder that allows the baserunners to advance.		they add more value than other players so they would help the team win more.
Batting.SecA	(BB + SB - CS + total bases - hits) / At-bats	New	Decrease, this is an advanced hitting statistic that measures efficiency of the batter. These players will affect winning more than the payroll.
Batting.PwrSpd	2*HR*SB/(HR+SB)	New	Increase, players and teams that have high power speed numbers are quite athletic and can perform well at two very different aspects of the game. This would make me think that a higher power speed number would make a team better.
Batting.WAR	Complex statistic that tries to calculate how many wins a player is better than a replacement level player using a plethora of different statistics.	New	Increase, this should be directly related to player value so even though players who have a high WAR command higher salaries, they impact winning more.
Pitching.SV	A save occurs when a new pitcher is substituted in to close out a win for the team.	Old	Increase, this is almost directly associated with wins so I believe that this would be strongly associated with efficiency.
Pitching.HR	Home Runs allowed by pitchers	Old	Decrease, as pitchers allow more home runs they would win less.
Pitching.BB	Walks allowed by pitchers	Old	Decrease, as pitchers allow more walks, teams win less.
Pitching.SO	Number of strikeouts given by pitchers.	Old	Increase, as pitchers give out more strikeouts the teams will win less.
Pitching.HBP	Number of times a pitcher hit a batter with a pitch.	Old	Decrease, pitchers with command issues are not typically that effective in general, so this would be a sign of a bad team.
Pitching.BK	A balk is when the pitcher begins to throw to home plate but stops before he completes his throwing motion.	Old	Decrease, as with HBP, pitchers with balk issues tend to not be effective in general so this could be a sign of bad pitching.

Pitching.WP	A wild pitch occurs when the pitcher throws the ball past the catcher.	Old	Decrease, as with Balks and HBP, this could be a sign of greater issues with the pitchers.
Pitching.ERAP	(League ERA[21] / ERA) * 100	New	Decrease, measures relationship of ERA with the league average so should allow teams to make informed decisions about payroll due to the knowledge of where the player ranks in relation to other pitchers.
Pitching.WHIP	(BB+Hits)/Innings Pitched[22]	Old	Decrease, as pitchers allow more hits and walks they perform worse, which leads to teams not winning as often.
Pitching.ERA	(Earned Runs[23] / IP) * 9 Calculates the average number of runs given up in a typical 9 inning game	Old	Decrease with a small magnitude, this is a good measure of performance, but I hypothesize it is over valued by teams.
Pitching.H9	9*Hits/IP Calculates number of hits given up in a typical game	Old	Decrease, as teams allow more hits they will perform worse in terms of winning games.
Pitching.HR9	9*HR/IP Calculates number of home runs given up in a typical game	Old	Decrease, as teams allow more home runs they will perform worse in terms of winning games.
Pitching.LOB	Number of runners that are on base when an inning ends. (Left on Base)	New	Increase with small magnitude, this means that more baserunners are left on at the end of the inning. This could be a sign that pitchers on the team perform better in stressful situations, but this also could be a sign of deeper issues that the pitchers have because they allowing a lot of baserunners.
Pitching.BAbip	Same as Batting.BAbip but this is the BAbip that pitchers allow	New	Decrease, for the same reasons as Batting.BAbip but for the opposite reasons.
Pitching.BB9	9*BB/IP Number of walks allowed in a typical game	Old	Decrease, as teams allow more walks they will perform worse in terms of winning games.

[21] ERA is calculated for the whole league, factoring in the different stadiums where baseball is played.
[22] IP is the abbreviated name
[23] An earned run is given to a pitcher when a run scores while he is pitching that is not due to a fielding mistake.

Pitching.SO9	9*SO/IP Number of Strikeouts by pitchers in a typical game	Old	Increase, as teams strikeout more batters, they will perform better in terms of winning games.
Pitching.FIP	((HR[24] * 13) + (3 * (BB[25] + HBP[26])) – (2 * SO[27])) / IP[28] + FIP.Constant[29]	New	Decrease, this is a pitching stat that measures performance without the impact of good defense. Aligns more closely with actual pitcher performance and will translate to more team success.
Pitching.SOpW	SO/BB	New	Increase, as teams strikeout more batters compared to walks there is more value added then when pitchers allow more walks than strikeouts.
Pitching.WAR	Same as Batting.WAR, but for Pitchers	New	Increase, for the same reason as Batting.WAR.
Fielding.DefEff	Percentage of balls put into play by a batter that end in an out.	New	Increase, this statistic shows how the team as a whole is performing defensively. This should be highly correlated with team success while not impacting overall payroll.
Fielding.Percentage	(Putouts[30] + Assists[31]) / (Putouts + Assists + Errors[32])[33]	Old	Small increase, though fielding percentage is a useful statistic when evaluating fielding performance, I hypothesize that teams over value it, but not enough to overcome the positive effect it has on team performance.
Fielding.A	See footnote 29	Old	Small increase, this is a good measure of how good a team is

[24] Home Runs allowed.
[25] Walks given up.
[26] When a batter is hit by a pitch, he is awarded first base rather than continuing the at-bat.
[27] Strikeouts occur when a pitcher throws three strikes during an at-bat. This can occur by a player swinging at any ball or by not swinging at a ball in the strike zone.
[28] Innings Pitched.
[29] All of the multipliers and the FIP Constant are determined by the league that scales this statistic to the familiar ERA scale.
[30] A putout is awarded when an out is completed by a player.
[31] An assist is awarded when a player contributes to an out in the field.
[32] An error is given when a player makes a mistake that keeps his team from achieving an out.
[33] For example, when a ground ball is hit to the shortstop and he throws it to the first baseman to get an out, the shortstop is awarded an assist and the first baseman is awarded a putout.

			at turning groundballs into outs, but this could be overvalued by teams.
Fielding.E	See footnote 30	Old	Decrease, teams that have more errors are bad at defense so a higher number would lead to teams not performing well.

Average Age

This is the average age of the players on the team. While the average age does not vary a lot, it still is an important characteristic of the roster make up. Players do not enter free agency, where teams can bid for them, until after their sixth season in the major leagues. Players who have not yet entered free agency are only allowed to negotiate with their team, which makes the contracts less expensive for the teams. When more teams are willing to spend the money on the player, the cost of the player's services increases. This means that younger players are typically cheaper than older players, which leads to the hypothesis that as age increases, efficiency will decrease because older players are more expensive than younger players for the same performance. Older players also begin to digress, so by the time a player is in their mid-to-late thirties they make the same as a mid-twenty-year-old but are performing at a much lower level. The dataset divided this into pitchers age and batters age, so there will be two variables in the regression regarding age in the regression called *Pitching.Age* and *Batting.Age.*

V. Summary Statistics & Correlation

Table 2 Contains summary statistics for all variables, including those used in regressions in Appendix 4

Statistics	N	Mean	Median	St. Dev	Min	Max
Batting.BatAge	1158	28.626	28.5	1.291	25.2	33.5
Batting.3B	1158	29.208	29	9.82	3	61
Batting.SB	1158	99.914	96	37.957	14	314
Batting.CS	1158	41.096	39	16.31	3	118
Batting.BB	1158	516.213	519	88.016	147	775
Batting.OPSP	1158	97.183	97	8.474	73	126
Batting.TB	1158	2208.674	2237	302.7	650	2832
Batting.GDP	1158	119.239	121	21.57	28	174
Batting.HBP	1158	51.372	51	17.793	9	116
Batting.SH	1158	46.705	46	23.849	0	119
Batting.SF	1158	43.57	44	10.094	7	75
Batting.rOBA	1158	0.332	0.332	0.016	0.278	0.379
Batting.BAbip	1158	0.295	0.294	0.012	0.245	0.332
Batting.ISO	1158	0.152	0.152	0.022	0.088	0.227
Batting.SOper	1158	0.182	0.176	0.033	0.113	0.273
Batting.BBper	1158	0.086	0.085	0.01	0.061	0.12
Batting.RSper	1158	0.304	0.3	0.02	0.24	0.37
Batting.SBper	1158	0.706	0.71	0.065	0.48	0.9
Batting.XBTper	1158	0.416	0.42	0.037	0.31	0.53
Batting.HRper	1158	0.027	0.027	0.006	0.01	0.051
Batting.WAR	1158	19.117	19	8.296	-6.7	50.4
Batting.SO	1158	1092.454	1077	219.363	440	1654
Batting.BA	1158	0.258	0.259	0.013	0.212	0.294
Batting.SecA	1158	0.259	0.257	0.029	0.183	0.349
Batting.PwrSpd	1158	117.277	117.3	28.762	24.3	210.5
Pitching.PAge	1158	28.574	28.5	1.388	24.5	34.2
Pitching.SV	1158	39.668	40	8.688	6	68
Pitching.SO	1158	1092.454	1076.5	219.566	393	1687
Pitching.HBP	1158	51.372	51	16.503	13	110
Pitching.BK	1158	7.061	6	6.288	0	76
Pitching.WP	1158	52.049	51	13.722	11	98
Pitching.ERAP	1158	101.64	101	10.527	77	146
Pitching.WHIP	1158	1.357	1.357	0.1	1.048	1.733
Pitching.HR9	1158	1.045	1	0.214	0.5	1.9
Pitching.LOB	1158	1102.634	1126	137.283	348	1338

Pitching.BAbip	1158	0.295	0.295	0.012	0.252	0.344
Pitching.ERA	1158	4.216	4.17	0.561	2.8	6.38
Pitching.BB9	1158	3.31	3.3	0.428	2.1	4.9
Pitching.SO9	1158	7.018	6.8	1.222	4.2	11
Pitching.WAR	1158	13.316	13.4	6.937	-10.4	36.5
Pitching.HR	1158	162.649	164	35.783	62	305
Pitching.BB	1158	516.213	521	84.896	145	784
Fielding.A	1158	1584.645	1625	229.963	442	2028
Fielding.E	1158	103.642	103	23.308	20	181
Pitching.FIP	1158	4.223	4.2	0.452	2.86	5.83
Pitching.SOpW	1158	2.164	2.095	0.517	1.16	3.98
Pitching.H9	1158	8.906	8.9	0.65	6.9	10.8
Fielding.DefEff	1158	0.693	0.693	0.012	0.642	0.732
Fielding.Fldper	1158	0.982	0.983	0.003	0.971	0.991
MVW	1158	2848155.339	1829138.978	6190361.958	97721.088	126295833
MVWP	1158	120.606	97.816	165.687	8.835	3502.1
DEPANA	1158	-4.595	-4.583	0.545	-8.161	-2.179

As seen in table 2, the performance team statistics are named by whether they are batting, pitching, or fielding statistics followed by the name of the statistic. There are some statistics that are named the same for batting and pitching, and they need to be differentiated for the model to function correctly. Additionally, this is useful for interpretability because it quickly shows how batting, pitching, and fielding interact with team success generally. Initially there were 70 variables, but this was brought down to 49 independent variables through correlation analysis for each regression. If two variables had more than 0.7 correlation with one another, one was removed within each regression. All these variables were then added to one large dataset. The correlation matrix was too large to include in this paper, and though there are some variables that have significant correlation (Corr > 0.7) with another, they were never included in the same regression. This is true except for those outside of batting and pitching variables that have a high correlation with one another. These are the most interesting relationships found through the

correlation matrix. Specifically, the high correlation between *Batting.SO* and *Pitching.SO;* this is interesting because one would imagine that these two variables should be independent from one another because they are measuring the level of success of two different parts of the team. In some of the regressions, they are both included because they should be independent from one another. A lot of the new era statistics are highly correlated with the old era statistics and one another because they build off the old era statistics. An example of this is the relationship between *Pitching.ERAP* and *Pitching.WAR* (0.775). These both measure pitching performance in relation to the league average so it is reasonable that they would be highly correlated with one another. There are also quite a few statistics that measure similar things, but are not correlated, such as *Fielding.Percentage* and *Fielding.DefEff*. These are both statistics that measure defensive performance, but they are almost completely uncorrelated (0.194). Overall, none of the regressions that this study ran used two highly correlated (Corr > 0.7) variables to ensure that results are useful and valid for interpretation.

VI. Regressions & Results

Reasoning for Regressions

For this research three main regressions will be built, additional models will be in Appendix 4. The goal is to compare the differences between the models and the coefficients to determine which variables explain the variation in the MWVP for a team. All regressions will include *Pitching.Age* and *Batting.Age* as a control and use a negative-log-transformation of MWVP as the dependent variable. The first model will include all the rate stats, the second will include all the old era statistics and be called the basic model, while the third will include all the new-era statistics and be called the advanced model. For the first model, any non-rate statistic was removed from the large list of variables that was previously discussed. Rate statistics are

defined as any variable that takes an outcome and divides it by the number of opportunities or a variable that is a percentage. For example, hits is a non-rate or count stat while batting average is a rate stat because it is hits divided by at-bats. For the second and third models, the first model was divided into the old (basic) and new (advanced) era statistics. Then each model underwent a correlation analysis that removed any highly correlated variables (Corr > 0.7). The age variables were also added to each model as a control for older players costing teams more in free agency. The goal for using these regressions is to determine if the new-era statistics are more useful in determining team efficiency than the old-era statistics.

Regression Results

Table 3 Regression table with estimates listed and the standard errors listed below in parenthesis.

	Dependent variable:		
		DEPANA	
	Rate	Basic	Advanced
	(1)	(2)	(3)
Batting.BatAge	-0.132***	-0.133***	-0.128***
	(0.012)	(0.012)	(0.012)
log(Batting.rOBA)	1.238		-0.384
	(1.198)		(0.751)
log(Batting.BAbip)	1.546*		2.249***
	(0.801)		(0.484)
log(Batting.BA)		3.872***	
		(0.378)	
log(Batting.HRper)	0.227	0.295***	
	(0.172)	(0.077)	
log(Batting.SOper)	0.367	0.622***	
	(0.237)	(0.154)	
log(Batting.BBper)	0.517***	0.786***	
	(0.180)	(0.118)	
log(Batting.RSper)	1.714***		1.183***
	(0.299)		(0.274)

log(Batting.SBper)	0.110	0.357**	
	(0.152)	(0.149)	
log(Batting.XBTper)	-0.055	0.323**	
	(0.160)	(0.154)	
log(Batting.SecA)			1.177***
			(0.237)
Batting.PwrSpd			0.0001
			(0.0005)
Batting.WAR	-0.001		0.004
	(0.003)		(0.003)
Pitching.PAge	-0.096***	-0.096***	-0.095***
	(0.010)	(0.011)	(0.011)
Pitching.ERA	-0.432***		
	(0.081)		
Pitching.H9		-0.273***	
		(0.033)	
Pitching.HR9	-0.019	-0.414***	
	(0.138)	(0.091)	
Pitching.BB9	-0.014	-0.199***	
	(0.043)	(0.034)	
Pitching.SO9	-0.092***	-0.098***	
	(0.023)	(0.021)	
Pitching.FIP			-0.296***
			(0.050)
log(Pitching.SOpW)			-0.159
			(0.107)
Pitching.WAR	0.006**		0.012***
	(0.003)		(0.003)
log(Fielding.DefEff)	0.770		6.230***
	(1.370)		(0.973)
log(Fielding.Fldper)	17.172***	15.141***	
	(5.182)	(4.785)	
Constant	13.002***	16.174***	10.487***
	(1.174)	(0.972)	(1.070)

Observations	1,158	1,158	1,158
R^2	0.422	0.380	0.391
Adjusted R^2	0.412	0.373	0.385
Residual Std. Error	0.418 (df = 1139)	0.432 (df = 1144)	0.428 (df = 1145)
F Statistic	46.109^{***} (df = 18; 1139)	53.863^{***} (df = 13; 1144)	61.367^{***} (df = 12; 1145)
Note:			$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$

VII. Interpretation of Results[34]

Batting.Age

Batting.Age is one of the two variables that is in all three main models. In all models there is a negative correlation with efficiency. For both regressions, it is significant at the 0.01 significance level and has a similar coefficient. The Rate model coefficient shows that when the average age of batters on a team increases by 1 year there is a 13.2% decrease in efficiency relative to the league average. The Basic coefficient shows that when the average age of batters increases by 1 there is a 13.3% decrease in efficiency in reference to the league average. The Advanced model shows that when the average age of batters increases by 1 there is a 12.8% decrease in efficiency. This follows the stated hypothesis for this variable because as players get older, their contracts are more expensive causing the team to need to pay more. This means that age has a strong positive correlation with the team's payroll, which is the numerator in the MWVP statistic.

[34] For the regressions, the dependent variable is a negative transformation of MWVP so a negative coefficient is a decrease in this negative variable but an increase in MWVP. For the interpretation, since an increase in MWVP is a decrease in efficiency, the interpretation will say that a negative coefficient is a decrease in efficiency by some percentage.

Pitching.Age

Pitching.Age is the other variable that is in all three models. As with B*atting.Age*, there exists a strong positive correlation with MWVP as it is statistically significant in all models at the 0.01 significance level. The Rate coefficient suggests that every increase of average pitcher age on a team by 1 year leads to efficiency decreasing by 9.6% in relation to the league average. The Basic coefficient shows that for every 1-year increase of average pitcher age on a team, efficiency decreases by 9.6% in relation to the league average. The Advanced model suggests that for every increase in average pitcher age by 1 year the team's efficiency decreases by 9.5% in relation to the league average. This statistic had the same hypothesis as the *Batting.Age* variable and these results support the hypothesis in the same way that is listed above.

These age results make sense because younger players are "team-controlled[35]" for the first six years that they are in the major leagues. Typically, this means that the players who are younger are cheaper, so the teams with a larger percentage of players typically would be spending less for their team to have more talent. This is the relationship that this variable is capturing, the length of service time and cost of the player to the team. Teams should look to develop players within their system so that the average MLB service time for the team would be lower, which would decrease payroll. The results also suggest that older pitchers have less of a decrease in efficiency than older batters. This is shown by the fact that the coefficients for *Batting.Age* are always larger than the coefficients for *Pitching.Age.* This could possibly mean that pitchers regress at a slower rate when looking through the lens of contributing to a winning

[35] Team controlled means that the player cannot negotiate with other teams for a contract.

team. This is quite interesting because typically, batters are viewed as regressing slower than pitchers due to the heightened risk of suffering a career ending injury as a pitcher.

Batting.BBper

Batting.BBper underwent a log-transformation and has a significant, positive relationship with efficiency. It is significant at the 0.01 level in both the Rate and Basic model. In the Rate model, the coefficient is 0.517, which means that a 1% increase in BBper leads to a 0.517% in efficiency relative to the league average. In the Basic model, the coefficient is 0.786, which means that a 1% increase in BBper leads to a 0.786% increase in efficiency. This means that as a team's batters walk more, the team is more efficient at using their resources.

Fielding.DefEff

Fielding.DefEff underwent a log-transformation and has a significant, positive relationship with efficiency. In the Rate model it is not significant, but it is significant at the 0.01 significance level with a coefficient of 6.230 in the Advanced model. This means that a 1% increase in a team's defensive efficiency leads to a 6.23% increase in the team's efficiency in relation to the league average. Practically, this means that as a team converts more balls in play to outs, their efficiency will increase. This result supports the hypothesis regarding this variable.

VIII. Conclusion

Overall, these models show that certain baseball statistics can be used to find how efficiently a team uses their resources. The most surprising result while comparing these models was that the *Fielding.Percentage* had a much larger coefficient than the variable *Fielding.DefEff.* These two statistics are measuring fielding in a similar way: the plays the team made divided by the plays the team could potentially make. The models had adjusted R^2 values of 0.412, 0.373,

and 0.385, which means that the Rate model variables explained 41.2% of the variation in team efficiency, the Basic model variables explained 37.3% of the variation in team efficiency, and the Advanced model variables explained 38.5% of the variation in team efficiency. This means that the Advanced model, the sabermetrics, explained the variation better than the basic rate statistics and while the direction hypotheses were quite accurate, the magnitude hypotheses were often quite surprising. This means that for the same set of MWVP, the variation in the advanced statistics better explain the variation in MWVP than the basic statistics. This supports the overarching hypothesis of this study that advanced, or sabermetrics statistics, are a better indicator of team efficiency and success. A unique branch off of this study would be to look at this from the player level using WAR and their salary. This model would be far more in depth and could provide teams with a strategy when trading for players.

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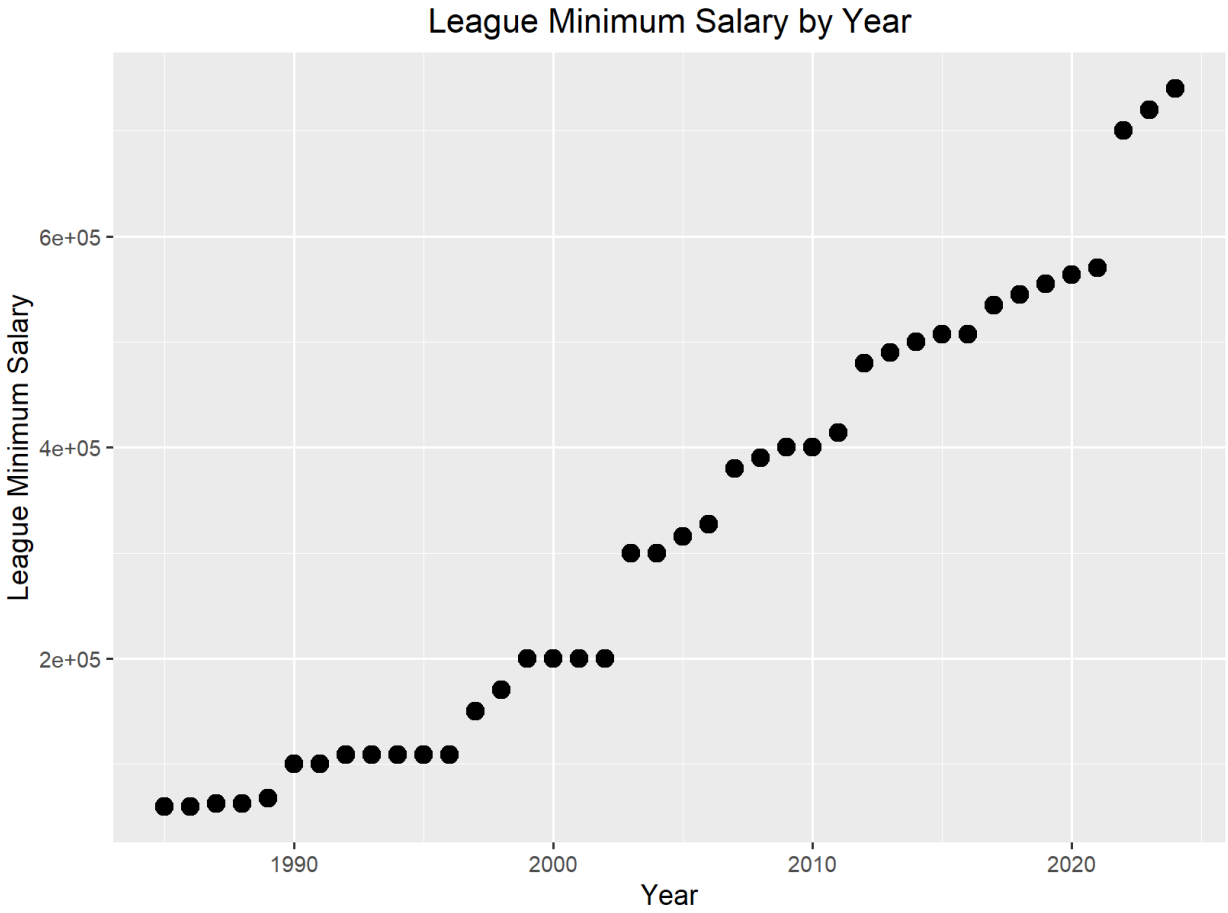
Appendix 1

League Minimum Contract Value

Figure 1 This graph shows the minimum salary for a major league player for each year included in the study. Each time there is a large jump in this value means that there is a new Collective Bargaining Agreement. ("MLB Minimum Salaries," n.d.; "Major League Baseball Minimum Wage | Baseball Almanac," n.d.)



Appendix 2

For the 2024 season the minimum payroll was $20,720,000 and the league average marginal win value was $4,612,652.36. Interestingly, The Cleveland Guardians and Houston Astros only played 161 games so their results were calculated using the 161 total games they played while all other teams played 162.

Table 4 Summary of 2024 MLB teams Marginal Win Value calculation.

Tm	W	WinPer	DiffWPer	WinDiff	Payroll	DiffPay	MWV	MWVP	nlMWVP
Arizona Diamondbacks	89	0.549	0.249	40.4	161,660,152	140,940,152	3488617.62	75.63	-4.326
Atlanta Braves	89	0.549	0.249	40.4	260,912,300	240,192,300	5945353.96	128.89	-4.859
Baltimore Orioles	91	0.562	0.262	42.4	117,347,715	96,627,715	2278955.54	49.41	-3.900
Boston Red Sox	81	0.500	0.200	32.4	160,834,775	140,114,775	4324530.09	93.75	-4.541
Chicago Cubs	83	0.512	0.212	34.4	223,048,500	202,328,500	5881642.44	127.51	-4.848
Chicago White Sox	41	0.253	-0.047	-7.6	130,170,125	109,450,125	109450125.00	2372.82	-7.772
Cincinnati Reds	77	0.475	0.175	28.4	98,323,334	77,603,334	2732511.76	59.24	-4.082
Cleveland Guardians	92	0.571	0.271	43.7	94,230,728	73,510,728	1682167.69	36.47	-3.596
Colorado Rockies	61	0.377	0.077	12.4	147,069,963	126,349,963	10189513.15	220.90	-5.398
Detroit Tigers	86	0.531	0.231	37.4	105,584,833	84,864,833	2269113.18	49.19	-3.896
Houston Astros	88	0.547	0.247	39.7	275,678,766	254,958,766	6422135.16	139.23	-4.936
Kansas City Royals	86	0.531	0.231	37.4	122,692,261	101,972,261	2726531.04	59.11	-4.079
Los Angeles Angels	63	0.389	0.089	14.4	167,637,049	146,917,049	10202572.85	221.19	-5.399
Los Angeles Dodgers	98	0.605	0.305	49.4	262,386,278	241,666,278	4892029.92	106.06	-4.664
Miami Marlins	62	0.383	0.083	13.4	109,992,600	89,272,600	6662134.33	144.43	-4.973
Milwaukee Brewers	93	0.574	0.274	44.4	115,142,460	94,422,460	2126631.98	46.10	-3.831
Minnesota Twins	82	0.506	0.206	33.4	135,762,591	115,042,591	3444388.95	74.67	-4.313
New York Mets	89	0.549	0.249	40.4	262,702,250	241,982,250	5989659.65	129.85	-4.866
New York Yankees	94	0.580	0.280	45.4	291,463,084	270,743,084	5963504.05	129.29	-4.862
Oakland Athletics	69	0.426	0.126	20.4	55,690,000	34,970,000	1714215.69	37.16	-3.615
Philadelphia Phillies	95	0.586	0.286	46.4	248,632,283	227,912,283	4911902.65	106.49	-4.668
Pittsburgh Pirates	76	0.469	0.169	27.4	85,226,500	64,506,500	2354251.83	51.04	-3.933
San Diego Padres	93	0.574	0.274	44.4	159,391,555	138,671,555	3123233.22	67.71	-4.215
San Francisco Giants	80	0.494	0.194	31.4	232,370,883	211,650,883	6740473.98	146.13	-4.984
Seattle Mariners	85	0.525	0.225	36.4	132,047,548	111,327,548	3058449.12	66.31	-4.194
St. Louis Cardinals	83	0.512	0.212	34.4	189,534,400	168,814,400	4907395.35	106.39	-4.667
Tampa Bay Rays	80	0.494	0.194	31.4	138,533,112	117,813,112	3752009.94	81.34	-4.399
Texas Rangers	78	0.481	0.181	29.4	261,471,834	240,751,834	8188837.89	177.53	-5.179
Toronto Blue Jays	74	0.457	0.157	25.4	225,628,287	204,908,287	8067255.39	174.89	-5.164
Washington Nationals	71	0.438	0.138	22.4	132,086,867	111,366,867	4971735.13	107.78	-4.680

For each team, the number of wins is divided by the total number of games played to get win percentage. This is then used to find how many more wins they have above the 0.300 threshold. This value is found in the WinDiff column above. For the payroll data, the league

average payroll mentioned above, $20,720,000, is subtracted from each team's payroll. This is found for each team in the DiffPay column. These two values are then used to find MWV by dividing the DiffPay by the WinDiff (DiffPay/WinDiff). These values are then divided by the aforementioned league average MWV, $4,612,652.36, to get the MWVP values for each team. Then the natural log is taken and is multiplied by -1, but this nlMWVP value is only used for the regressions and has very little interpretability on its own.

The Cleveland Guardians have a MWVP value of 36.47 and the Oakland Athletics have a MWVP of 37.16, but these teams 2024 seasons could not have been more different. Cleveland won 23 games and made the playoffs, while Oakland was the fifth worst team and are in a full scale rebuild. The similarity is that they both use their resources efficiently and win at a higher rate given their payroll. These teams can then be compared with the Los Angeles Angels and the Colorado Rockies who have a 221.19 and 220.90 MWVP, respectively. Both of these teams did not have a great year, they did not win at all, but they spent a lot of money on their players. These are the teams that are extremely bad at using their resources.

Appendix 3

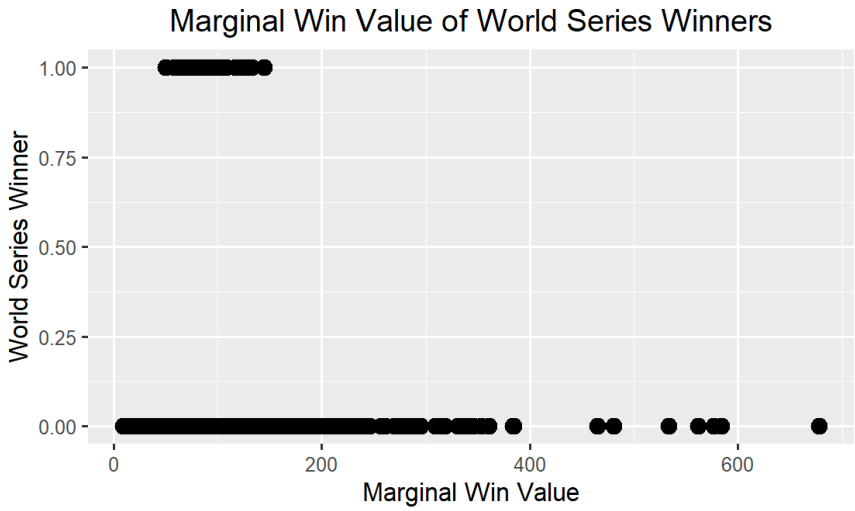


Figure 2 This shows the relationship between World Series winning teams (1) and what their MWVP was compared to the teams that did not win the World Series (0)

As can be seen in the figure above, World Series Winners from the past 40 years come from a specific range of MWVP values. The only teams that have ultra-low values of MWVP are those that do not spend money, whereas the teams that have extremely high values of MWVP are those that do not win often but spend a lot of money. This graph shows that to win the World Series teams must find a happy medium between being efficient and spending money when necessary.

Appendix 4

Table 5 Extra regressions using count, full, batting, and pitching statistics

Comparing Extra Models

	Dependent variable:			
	DEPANA			
	Full	Count	Batting	Pitching
	(1)	(2)	(3)	(4)
Batting.BatAge	-0.133***	-0.138***	-0.149***	
	(0.012)	(0.012)	(0.013)	
Batting.3B	-0.001	0.002	0.003*	
	(0.002)	(0.002)	(0.002)	
Batting.SB	0.0004	0.0004	0.002	
	(0.001)	(0.0005)	(0.001)	
Batting.CS	0.0001	-0.001	-0.003	
	(0.003)	(0.001)	(0.003)	
Batting.BB	-0.002	0.001***	-0.002	
	(0.001)	(0.0002)	(0.001)	
log(Batting.OPSP)	0.031		1.537***	
	(0.384)		(0.252)	
Batting.SO		0.0001		
		(0.0001)		
Batting.TB	-0.0002	0.0001	0.0003	
	(0.0003)	(0.0001)	(0.0003)	
Batting.GDP	-0.001	-0.003***	-0.001	
	(0.001)	(0.001)	(0.001)	
Batting.HBP	0.004***	0.004***	0.005***	
	(0.001)	(0.001)	(0.001)	
Batting.SH	0.001	0.003***	0.004***	
	(0.001)	(0.001)	(0.001)	
Batting.SF	-0.001	-0.0003	0.002	
	(0.002)	(0.002)	(0.002)	
log(Batting.rOBA)	-2.831**		-1.271	
	(1.398)		(1.425)	
log(Batting.BAbip)	3.498***		0.235	
	(0.861)		(0.832)	
log(Batting.ISO)	1.295***		-0.185	
	(0.436)		(0.420)	
log(Batting.SOper)	-0.521*		-0.005	
	(0.269)		(0.288)	
log(Batting.BBper)	1.687***		1.481**	
	(0.624)		(0.725)	
log(Batting.RSper)	1.603***		1.082***	
	(0.309)		(0.364)	
log(Batting.SBper)	0.085		-0.176	
	(0.347)		(0.415)	
log(Batting.XBTper)	-0.130		0.050	
	(0.162)		(0.191)	
Batting.WAR		0.017***		
		(0.003)		

Pitching.PAge	-0.099***	-0.103***		-0.125***
	(0.010)	(0.010)		(0.010)
Pitching.SV	0.018***	0.018***		0.022***
	(0.002)	(0.002)		(0.002)
Pitching.HR		-0.001**		
		(0.001)		
Pitching.BB		-0.0004		
		(0.0002)		
Pitching.SO	-0.001***	-0.001***		-0.0003**
	(0.0002)	(0.0001)		(0.0002)
Pitching.HBP	-0.003**	-0.0002		0.002
	(0.001)	(0.001)		(0.001)
Pitching.BK	-0.004*	-0.006**		-0.004*
	(0.002)	(0.002)		(0.002)
Pitching.WP	0.001	0.001		0.001
	(0.001)	(0.001)		(0.001)
log(Pitching.ERAP)	0.367			1.377***
	(0.307)			(0.193)
Pitching.WHIP	-1.834***			-0.353
	(0.568)			(0.460)
Pitching.HR9	-0.249**			0.003
	(0.121)			(0.114)
Pitching.LOB	0.001***			-0.0004*
	(0.0005)			(0.0002)
log(Pitching.BAbip)	-0.333			-0.077
	(0.552)			(0.596)
log(Fielding.Fldper)	11.092**			
	(4.668)			
Pitching.WAR		0.020***		
		(0.003)		
Fielding.A		-0.0004***		
		(0.0001)		
Fielding.E		-0.001*		
		(0.001)		
Constant	11.111**	2.019***	-4.235**	-7.173***
	(4.792)	(0.376)	(1.915)	(1.533)
Observations	1,158	1,158	1,158	1,158
R^2	0.483	0.442	0.246	0.292
Adjusted R^2	0.469	0.430	0.233	0.285
Residual Std. Error	0.397 (df = 1126)	0.412 (df = 1134)	0.478 (df = 1138)	0.461 (df = 1146)
F Statistic	33.995*** (df = 31; 1126)	38.986*** (df = 23; 1134)	19.538*** (df = 19; 1138)	42.922*** (df = 11; 1146)
Note:				*p<0.1; **p<0.05; ***p<0.01

Table 6 Comparing Advanced Regression for five-year periods.

Comparing Advanced Model for 5-year Intervals

	Dependent variable:			
	DEPANA			
	1985-1989	1990-1994	1995-1999	2000-2004
	(1)	(2)	(3)	(4)
Batting.BatAge	-0.085***	-0.139***	-0.146***	-0.147***
	(0.026)	(0.029)	(0.031)	(0.031)
log(Batting.rOBA)	1.740	-3.026	1.932	2.685
	(2.496)	(2.659)	(3.138)	(3.508)
log(Batting.BAbip)	2.881**	5.019***	-0.345	2.992*
	(1.326)	(1.190)	(1.692)	(1.687)
log(Batting.RSper)	2.922***	0.997	2.054**	1.210
	(0.661)	(0.660)	(0.880)	(0.912)
log(Batting.SecA)	0.914	1.369**	0.174	0.965
	(0.731)	(0.656)	(0.974)	(0.912)
Batting.PwrSpd	-0.001	0.0002	-0.0004	-0.0004
	(0.002)	(0.001)	(0.002)	(0.002)
Batting.WAR	-0.015	0.014	-0.004	-0.001
	(0.012)	(0.008)	(0.009)	(0.010)
Pitching.PAge	-0.040**	-0.079***	-0.034	-0.142***
	(0.020)	(0.023)	(0.029)	(0.029)
Pitching.FIP	-0.213	-0.115	-0.337*	-0.450**
	(0.174)	(0.140)	(0.183)	(0.189)
log(Pitching.SOpW)	0.169	-0.383	0.054	0.075
	(0.306)	(0.274)	(0.426)	(0.357)
Pitching.WAR	0.010	0.008	0.012*	0.014
	(0.009)	(0.007)	(0.007)	(0.009)
log(Fielding.DefEff)	15.519***	6.834**	4.852*	11.564***
	(3.498)	(2.685)	(2.610)	(3.316)
Constant	15.795***	10.411***	8.147**	19.108***
	(4.442)	(3.252)	(3.435)	(4.037)
Observations	130	134	144	150
R^2	0.622	0.555	0.389	0.570
Adjusted R^2	0.584	0.510	0.333	0.532
Residual Std. Error	0.314 (df = 117)	0.306 (df = 121)	0.378 (df = 131)	0.414 (df = 137)
F Statistic	16.074*** (df = 12; 117)	12.553*** (df = 12; 121)	6.947*** (df = 12; 131)	15.114*** (df = 12; 137)
Note:				*p<0.1; **p<0.05; ***p<0.01

Table 7 Comparing Advanced Regression for five-year periods

Comparing Advanced Model for 5-year Intervals

	Dependent variable:			
	DEPANA			
	2005-2009	2010-2014	2015-2019	2020-2024
	(1)	(2)	(3)	(4)
Batting.BatAge	-0.097***	-0.203***	-0.119**	-0.215***
	(0.031)	(0.031)	(0.050)	(0.046)
log(Batting.rOBA)	-11.037***	-2.606	-2.377	-3.655
	(3.417)	(2.241)	(3.893)	(2.895)
log(Batting.BAbip)	3.870**	-0.409	1.593	1.785
	(1.654)	(1.398)	(1.929)	(1.662)
log(Batting.RSper)	1.239	1.538**	1.325	2.030**
	(0.786)	(0.711)	(0.951)	(0.868)
log(Batting.SecA)	3.411***	2.096***	1.136	1.910*
	(0.781)	(0.671)	(0.941)	(0.981)
Batting.PwrSpd	-0.001	-0.00003	0.001	-0.001
	(0.001)	(0.002)	(0.002)	(0.002)
Batting.WAR	0.020*	0.002	0.016	0.010
	(0.010)	(0.009)	(0.013)	(0.009)
Pitching.PAge	-0.104***	-0.125***	-0.145***	-0.180***
	(0.027)	(0.032)	(0.046)	(0.043)
Pitching.FIP	-0.045	-0.635***	-0.737***	-0.643***
	(0.190)	(0.180)	(0.201)	(0.223)
log(Pitching.SOpW)	0.365	-0.384	-1.007**	0.023
	(0.338)	(0.345)	(0.476)	(0.513)
Pitching.WAR	0.014	0.006	0.007	0.007
	(0.009)	(0.008)	(0.012)	(0.010)
log(Fielding.DefEff)	0.005	7.251**	6.681*	0.902
	(3.100)	(2.982)	(3.474)	(2.792)
Constant	-0.515	11.607***	11.297**	12.407***
	(4.109)	(3.461)	(4.678)	(3.359)
Observations	150	150	150	150
R^2	0.464	0.568	0.380	0.491
Adjusted R^2	0.417	0.531	0.326	0.446
Residual Std. Error (df = 137)	0.382	0.356	0.500	0.507
F Statistic (df = 12; 137)	9.897***	15.032***	7.012***	11.002***
Note:				*p<0.1; **p<0.05; ***p<0.01

Table 8 Comparing models only using full seasons (not 1994, 1995, or 2020)

Comparing Main Models Using Full Seasons

	Dependent variable:		
		DEPANA	
	Rate	Basic	Advanced
	(1)	(2)	(3)
Batting.BatAge	-0.133***	-0.133***	-0.126***
	(0.013)	(0.013)	(0.012)
log(Batting.rOBA)	1.433		-0.235
	(1.321)		(0.839)
log(Batting.BAbip)	1.800**		2.203***
	(0.870)		(0.514)
log(Batting.BA)		4.046***	
		(0.406)	
log(Batting.HRper)	0.312*	0.323***	
	(0.184)	(0.081)	
log(Batting.SOper)	0.262	0.617***	
	(0.253)	(0.163)	
log(Batting.BBper)	0.531***	0.775***	
	(0.190)	(0.124)	
log(Batting.RSper)	1.821***		1.169***
	(0.323)		(0.295)
log(Batting.SBper)	0.110	0.332**	
	(0.163)	(0.159)	
log(Batting.XBTper)	-0.047	0.322*	
	(0.172)	(0.165)	
log(Batting.SecA)			1.136***
			(0.252)
Batting.PwrSpd			0.0002
			(0.001)
Batting.WAR	-0.005		0.005
	(0.004)		(0.004)
Pitching.PAge	-0.100***	-0.102***	-0.100***
	(0.011)	(0.011)	(0.011)
Pitching.ERA	-0.432***		
	(0.092)		
Pitching.H9		-0.294***	
		(0.035)	

Pitching.HR9	-0.084	-0.396***	
	(0.147)	(0.096)	
Pitching.BB9	-0.028	-0.196***	
	(0.047)	(0.036)	
Pitching.SO9	-0.077***	-0.101***	
	(0.025)	(0.022)	
Pitching.FIP			-0.280***
			(0.054)
log(Pitching.SOpW)			-0.138
			(0.115)
Pitching.WAR	0.003		0.013***
	(0.004)		(0.003)
log(Fielding.DefEff)	1.929		6.540***
	(1.517)		(1.160)
log(Fielding.Fldper)	17.057***	14.160***	
	(5.597)	(5.056)	
Constant	14.523***	16.794***	10.658***
	(1.524)	(1.027)	(1.390)
Observations	1,072	1,072	1,072
R^2	0.421	0.384	0.394
Adjusted R^2	0.411	0.376	0.388
Residual Std. Error	0.421 (df = 1053)	0.434 (df = 1058)	0.430 (df = 1059)
F Statistic	42.564*** (df = 18; 1053)	50.662*** (df = 13; 1058)	57.483*** (df = 12; 1059)
Note:			*p<0.1; **p<0.05; ***p<0.01

Table 9 Comparing models only using full seasons (not 1994, 1995, or 2020)

	Dependent variable:			
	DEPANA			
	Full	Count	Batting	Pitching
	(1)	(2)	(3)	(4)
Batting.BatAge	-0.133***	-0.137***	-0.148***	
	(0.012)	(0.012)	(0.013)	
Batting.3B	-0.001	0.003*	0.002	
	(0.002)	(0.002)	(0.002)	
Batting.SB	0.001	0.001	0.002	
	(0.001)	(0.001)	(0.001)	
Batting.CS	-0.001	-0.001	-0.004	
	(0.003)	(0.001)	(0.003)	
Batting.BB	-0.002	0.001***	-0.003	
	(0.002)	(0.0002)	(0.002)	
log(Batting.OPSP)	-0.044		1.620***	
	(0.409)		(0.266)	
Batting.SO		0.0003*		
		(0.0001)		
Batting.TB	-0.001	0.0002	-0.002***	
	(0.001)	(0.0001)	(0.001)	
Batting.GDP	-0.001	-0.002*	-0.002**	
	(0.001)	(0.001)	(0.001)	
Batting.HBP	0.004***	0.004***	0.004***	
	(0.001)	(0.001)	(0.001)	
Batting.SH	0.001	0.003***	0.003***	
	(0.001)	(0.001)	(0.001)	
Batting.SF	-0.001	0.001	0.002	
	(0.002)	(0.002)	(0.002)	
log(Batting.rOBA)	-2.345		0.444	
	(1.518)		(1.551)	
log(Batting.BAbip)	4.227***		2.927***	
	(1.044)		(1.113)	
log(Batting.ISO)	1.941***		1.581**	
	(0.572)		(0.626)	
log(Batting.SOper)	-0.773**		-0.701**	
	(0.311)		(0.345)	
log(Batting.BBper)	1.624		1.945*	
	(0.990)		(1.158)	
log(Batting.RSper)	1.680***		1.029***	
	(0.326)		(0.380)	
log(Batting.SBper)	-0.081		-0.273	
	(0.404)		(0.479)	
log(Batting.XBTper)	-0.166		-0.033	
	(0.173)		(0.202)	
Batting.WAR		0.017***		
		(0.003)		
Pitching.PAge	-0.104***	-0.105***		-0.131***
	(0.011)	(0.011)		(0.011)
Pitching.SV	0.017***	0.019***		0.023***
	(0.002)	(0.002)		(0.002)

Pitching.HR		-0.001		
		(0.001)		
Pitching.BB		-0.0002		
		(0.0003)		
Pitching.SO	-0.001***	-0.0004***		-0.0003*
	(0.0002)	(0.0001)		(0.0002)
Pitching.HBP	-0.003***	-0.001		0.001
	(0.001)	(0.001)		(0.001)
Pitching.BK	-0.004*	-0.004*		-0.004*
	(0.002)	(0.002)		(0.002)
Pitching.WP	0.001	0.0004		0.001
	(0.001)	(0.001)		(0.001)
log(Pitching.ERAP)	0.126			1.211***
	(0.331)			(0.209)
Pitching.WHIP	-2.310***			-0.988
	(0.653)			(0.605)
Pitching.HR9	-0.223*			0.133
	(0.132)			(0.133)
Pitching.LOB	0.002***			0.001
	(0.001)			(0.001)
log(Pitching.BAbip)	-0.441			-0.357
	(0.602)			(0.651)
log(Fielding.Fldper)	11.257**			
	(4.905)			
Pitching.WAR		0.019***		
		(0.003)		
Fielding.A		-0.0001		
		(0.0002)		
Fielding.E		-0.001		
		(0.001)		
Constant	16.228***	0.597	9.064**	-7.104***
	(5.820)	(0.578)	(4.482)	(1.674)
Observations	1,072	1,072	1,072	1,072
R^2	0.488	0.453	0.264	0.294
Adjusted R^2	0.472	0.441	0.250	0.286
Residual Std. Error	0.399 (df = 1040)	0.411 (df = 1048)	0.476 (df = 1052)	0.464 (df = 1060)
F Statistic	31.941*** (df = 31; 1040)	37.732*** (df = 23; 1048)	19.823*** (df = 19; 1052)	40.073*** (df = 11; 1060)
Note:				*p<0.1; **p<0.05; ***p<0.01